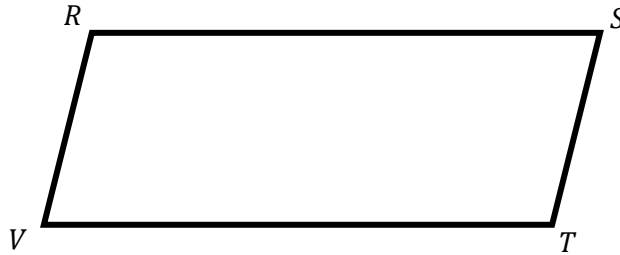


Geometry
Unit 8
Lesson 2 Practice

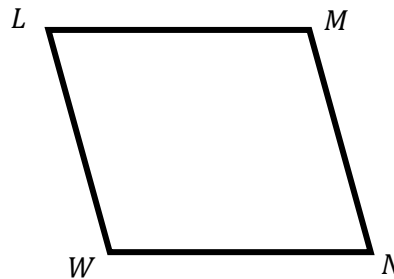
Name **Answer Key** _____
Date _____
Period _____

Parallelogram $RSTV$ is shown where $m\angle S = (4x - 8)^\circ$, $m\angle T = 144^\circ$, and $m\angle V = 36^\circ$.



1. I can conclude ...
 $m\angle R = 144^\circ$
2. I know this because ...
Opposite angles in a parallelogram are congruent and have equal measure
3. Since $RSTV$ is a parallelogram, what is the value of x ?
 $x = 11$

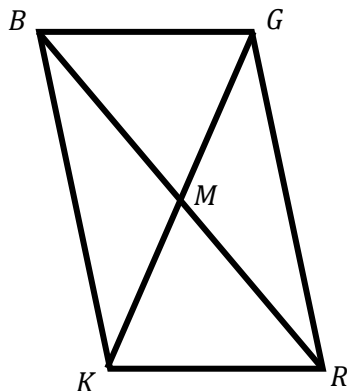
Quadrilateral $LMNW$ is shown where $m\angle M = (14x - 8)^\circ$, $m\angle N = (16y - 4)^\circ$, $m\angle W = (5x + 64)^\circ$, and $m\angle L = (12y + 16)^\circ$.



4. What value of x will make $LMNW$ a parallelogram?
 $x = 8$
5. What value of y will make $LMNW$ a parallelogram?
 $y = 5$
6. Given that $LMNW$ is a parallelogram, what is the $m\angle M$?
 $m\angle M = 104^\circ$

Quadrilateral $BGRK$ is shown with diagonals $\overline{BR}$ and $\overline{GK}$ that intersect at point M .

$MB = 8z - 6$, $MR = 32$, $MG = 28$, and $MK = 3t + 3$, where $t = 8$.



Charlene and Jordan are discussing whether $BGRK$ is a parallelogram. Charlene says $BGRK$ is a parallelogram and Jordan says $BGRK$ is not a parallelogram.

7. Who is correct?

Jordan is correct. Diagonals of a parallelogram bisect each other which means that $KM = MG$. Using the given information $KM = 3(8) + 3 = 27$ and $MG = 28$. Since $KM \neq MG$, the quadrilateral $BGRK$ is not a parallelogram.

8. Which property of parallelograms was used to determine if $BGRK$ is a parallelogram?

Diagonals of a parallelogram bisect each other

9. Given that $BGRK$ is a parallelogram, what is the value of z ?

It is not possible to determine the value of z because $BGRK$ is not a parallelogram so there is not enough information to identify relationships with lengths of sides and diagonals of this quadrilateral.

10. What is the length of $\overline{MK}$?

$$KM = 3(8) + 3 = 27$$